Hi, you should behave as a professional scribe generator. Create a scribe as a LaTeX code of this file "L21_L22.pdf" file. Make the scribe keeping in mind that I need to reference this material for preparing for my exam. It should be precise and accurate. Keep in mind that you should not change the content in the file. Also add whatever formulas are required for the following lecture. Keep the content accurate. Follow the lecture's structure and scope. Don't add the content from the reference file and also do not add anything new yourself. Take the file "group_6_ITS_project_scribe (1).pdf" as the reference for the intended scribe's format. Make it possible so that it can be directly converted to LaTeX. For now just give me the LaTeX code for this file.

L26_Stocha...SS_Ergodic
PDF

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Code snippet

```latex
\documentclass[11pt]{article}
\usepackage[utf8]{inputenc}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath, amssymb, amsfonts}
\usepackage{enumitem}
\usepackage{graphicx}
\usepackage{hyperref}

\title{\textbf{Lecture 26: Stochastic Processes Part 1} \\ \large St
\author{Professional Scribe}
\date{April 14, 2026}

\begin{document}

\maketitle

\section{Introduction to Stochastic Processes}
A \textbf{stochastic process} $X(\zeta, t)$ is defined as an entire

\subsection{Key Concepts}
\begin{itemize}
    \item \textbf{Realization:} A specific measured waveform from an
    \item \textbf{Random Variable (R.V.):} If the process is "frozen
    \item \textbf{Multiple Variables:} By evaluating the process at
\end{itemize}

\section{Process Characterization}
Stochastic processes are characterized by their Probability Density
\begin{itemize}
    \item \textbf{1st Order:} Describes the R.V. at a single time $t
    \item \textbf{2nd Order:} Describes the joint probability across
    \item \textbf{$n^{th}$ Order:} Provides a complete statistical c
\end{itemize}

\section{Statistical Moments}

\subsection{Mean Function}
The mean is the expected value of the random variable at a given tim
\begin{equation}
    \mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x \cdot f_X(x, t) \
\end{equation}
Generally, the mean changes as a function of time[cite: 172].

\subsection{Autocorrelation Function}
Measures the relationship between the process sampled at two differe
\begin{equation}
    R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]
\end{equation}
\textbf{Properties:}
\begin{itemize}
```

```latex
    \item \textbf{Conjugate Symmetry:} $R_{XX}(t_1, t_2) = R_{XX}^*(
    \item \textbf{Positive Definiteness:} The function forms a posit
    $\sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j^* R_{XX}(t_i, t_j) \ge 0$[
\end{itemize}

\subsection{Autocovariance Function}
Measures the relationship between the deviations of the process from
\begin{equation}
    C_{XX}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X^*(t_2) - \mu_X^*(t_
\end{equation}
It is related to autocorrelation by: $C_{XX}(t_1, t_2) = R_{XX}(t_1,

\section{Stationary Stochastic Processes}

\subsection{Strict Sense Stationary (SSS)}
A process is SSS if its entire joint distribution is invariant under
\begin{equation}
    f_X(x_1, \dots, x_n; t_1, \dots, t_n) = f_X(x_1, \dots, x_n; t_1
\end{equation}
For an SSS process, the mean $\mu_X$ is constant and the autocorrela

\subsection{Wide Sense Stationary (WSS)}
WSS is a weaker condition often used in practice[cite: 483]. A proce
\begin{enumerate}
    \item \textbf{Constant Mean:} $E[X(t)] = \mu_X = \text{constant}
    \item \textbf{Lag-Dependent Autocorrelation:} $E[X(t_1)X^*(t_2)]
\end{enumerate}
\textit{Note: SSS implies WSS, but WSS does not necessarily imply SS

\section{Ergodic Process}
A process is \textbf{ergodic} if its time averages are equal to its
\begin{itemize}
    \item \textbf{Time Average (Mean):} $\langle X(t) \rangle = \lim
    \item \textbf{Time-Average Autocorrelation:} $R_{XX}(\tau) = \li
\end{itemize}
In an ergodic process, a single realization can represent the entire

\section{Summary of Mathematical Tools}
\begin{table}[h]
\centering
\begin{tabular}{|l|l|}
\hline
\textbf{Function} & \textbf{Formula} \\ \hline
Mean Function & $\mu_X(t) = E[X(t)]$ [cite: 635] \\ \hline
Autocorrelation & $R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$ [cite: 646] \
Autocovariance & $C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\m
Cross-Correlation & $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$ [cite: 666]
\end{tabular}
\end{table}
```